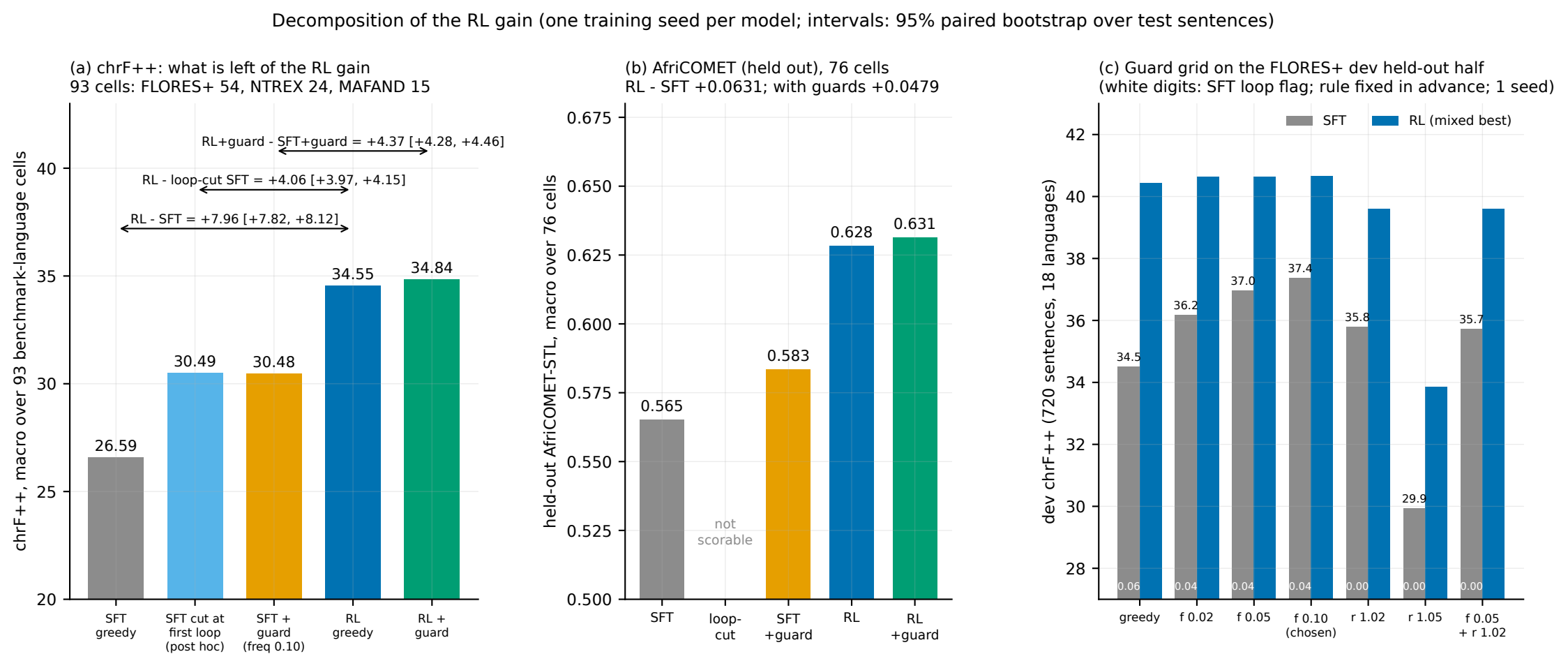
Decomposition of the RL gain (one training seed per model; intervals: 95% paired bootstrap over test sentences)
(a) chrF++: what is left of the RL gain
93 cells: FLORES+ 54, NTREX 24, MAFAND 15
RL+guard - SFT+guard = +4.37 [+4.28, +4.46]
RL - loop-cut SFT = +4.06 [+3.97, +4.15]
RL - SFT = +7.96 [+7.82, +8.12]
chrF++, macro over 93 benchmark-language cells
26.59
30.49
30.48
34.55
34.84
SFT greedy
SFT cut at first loop (post hoc)
SFT + guard (freq 0.10)
RL greedy
RL + guard
(b) AfriCOMET (held out), 76 cells
RL - SFT +0.0631; with guards +0.0479
held-out AfriCOMET-STL, macro over 76 cells
0.565
not scorable
0.583
0.628
0.631
SFT
loop-cut
SFT +guard
RL
RL +guard
(c) Guard grid on the FLORES+ dev held-out half
(white digits: SFT loop flag; rule fixed in advance; 1 seed)
SFT
RL (mixed best)
dev chrF++ (720 sentences, 18 languages)
34.5
36.2
37.0
37.4
35.8
29.9
35.7
0.06
0.04
0.04
0.04
0.00
0.00
0.00
greedy
f 0.02
f 0.05
f 0.10 (chosen)
r 1.02
r 1.05
f 0.05 + r 1.02